

Weekly Deliverable, Weeks 5–6: Lagrangian Decomposition and Dual Prices for the Mini Water–Energy Problem

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June 18, 2026

1 The Mini Water–Energy Problem

1.1 Two operators and one boundary

Two operators schedule over a day-ahead horizon $\mathcal{T} = \{1, \dots, T\}$, $T = 6$, at hourly resolution. A *water operator* pumps water into an elevated tank and serves a known demand profile; an *energy operator* procures electricity at the day-ahead price and serves the *water operator* to meet pump load that the water operator’s schedule creates. The two systems touch at a single electrical boundary of finite capacity. Table 1 collects the notation.

Table 1: Notation for the mini problem.

Symbol	Description	Unit
Sets		
$\mathcal{T}$	Set of time periods (hours), $\mathcal{T} = \{1, \dots, T\}$	–
Decision Variables		
q_t	Pumped water volume in hour t	m^3
s_t	Tank volume level at the end of hour t	m^3
$E^{\max}$	Peak electrical load at hour t	kWh
Parameters		
$\bar{Q}$	Maximum pump capacity	m^3
$S^{\min}, S^{\max}$	Minimum and maximum tank level bounds	m^3
s_0	Initial tank volume level	m^3
d_t^w	Water demand in hour t	m^3
κ	Specific pumping energy coefficient	kWh/m^3
$\bar{E}$	Boundary electricity import capacity limit	kWh
c_t	Day-ahead procurement price in hour t	$\$/\text{kWh}$

1.2 Centralized Coordination

The case where one operator owns everything.

$$\min_{q, s} \sum_{t \in \mathcal{T}} c_t e_t + \alpha \max_{t \in \mathcal{T}}(e_t) \quad (1)$$

$$\text{s.t. (water)} \quad s_t = s_{t-1} + q_t - d_t^w, \quad t \in \mathcal{T}, \quad (2)$$

$$S^{\min} \leq s_t \leq S^{\max}, \quad t \in \mathcal{T}, \quad (3)$$

$$0 \leq q_t \leq \bar{Q}, \quad t \in \mathcal{T}, \quad (4)$$

$$s_T \geq s_0 \quad (5)$$

$$\text{(energy)} \quad e_t = \kappa q_t, \quad t \in \mathcal{T}, ** \quad (6)$$

$$0 \leq e_t \leq \bar{E}, \quad t \in \mathcal{T}, \quad (7)$$

**** Coupling Constraint**

1.3 Decentralized Uncoordinated

In this case we look at the problem from the water operator side (i.e. the retailer in the news vendor model setup). Hence in this case the cost of the electricity will be production cost with marginal profit $p_t = c_t + \text{marginal}$. Other than the setup remain as the centralized problem. The program reads

$$\min_{q, s} \sum_{t \in \mathcal{T}} p_t e_t \quad (8)$$

$$\text{s.t. (water)} \quad s_t = s_{t-1} + q_t - d_t^w, \quad t \in \mathcal{T}, \quad (9)$$

$$S^{\min} \leq s_t \leq S^{\max}, \quad t \in \mathcal{T}, \quad (10)$$

$$0 \leq q_t \leq \bar{Q}, \quad t \in \mathcal{T}, \quad (11)$$

$$s_T \geq s_0 \quad (12)$$

$$\text{(energy)} \quad e_t = \kappa q_t, \quad t \in \mathcal{T}, ** \quad (13)$$

$$0 \leq e_t \leq \bar{E}, \quad t \in \mathcal{T}, \quad (14)$$

**** Coupling Constraint**

1.4 Decentralized Coordinated

In this case we look at the problem from the water operator side (i.e. the retailer in the news vendor model setup). Hence in this case the cost of the electricity will be production cost with marginal profit $p_t = c_t + \text{marginal}$. Other than the setup remain as the centralized problem. The program

reads

$$\min_{q, s, E^{\max}} \sum_{t \in \mathcal{T}} c_t q_t + \lambda E^{\max} \quad (15)$$

$$\text{s.t. (water)} \quad s_t = s_{t-1} + q_t - d_t^w, \quad t \in \mathcal{T}, \quad (16)$$

$$S^{\min} \leq s_t \leq S^{\max}, \quad t \in \mathcal{T}, \quad (17)$$

$$0 \leq q_t \leq \bar{Q}, \quad t \in \mathcal{T}, \quad (18)$$

$$s_T \geq s_0 \quad (19)$$

$$\text{(energy)} \quad e_t = \kappa q_t, \quad t \in \mathcal{T}, ** \quad (20)$$

$$0 \leq e_t \leq \bar{E}, \quad t \in \mathcal{T}, \quad (21)$$

**** Coupling Constraint**

1.5 Case study

In this section we will compare the three coordination scheme through the following problem setup presented in table 2.

Table 2: Parameter values for the centralized coordination model.

Symbol	Description	Value
$\mathcal{T}$	Time horizon index	$\{1, 2, 3, 4, 5, 6\}$
d_t^w	Water demand	2 m ³ (for every hour t)
κ	Specific pumping energy	1 kWh/m ³
$\bar{Q}$	Pump capacity ($q^{\max}$)	5 m ³ /h
$S^{\max}$	Maximum tank level limit	4 m ³
s_0	Initial tank level	0 m ³
c_t	Day-ahead procurement price	$\{1, 1, 2, 4, 4, 2\}$ \$/kWh
α	Peak cost coefficient	\$3

Two features of (??) deserve emphasis. First, the water side carries no cost of its own: the cost of pumping *is* the cost of the electricity that pumping consumes, and that cost appears once, on the energy side. This is the coupling decision identified earlier in the thesis work—removing the exogenous pump-electricity cost from the water objective and letting the pump load enter the energy balance instead—and (??) is its smallest expression. Second, the boundary balance is the *only* constraint involving both operators’ variables; everything else is private. The problem is therefore exactly of the form that Palomar and Chiang [4] identify as a problem with a coupling constraint, the form for which dual decomposition is the natural instrument.

References

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